

The localization of torsion to a single CRT block

A self-contained proof that any torsion in the integral cohomology
of a Fermat cell is confined to one computable block

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Abstract

This note isolates and proves, in full and on its own terms, the one rigorously established theorem underlying the author's computational campaign on the integral Hodge conjecture for Fermat varieties: *any torsion in the integral cohomology module of a Fermat cell is localized to a single block of the Chinese Remainder Theorem (CRT) splitting, namely the unique block whose rank drops modulo p* . The proof rests on one structural fact that holds by construction—the CRT idempotents are polynomials in the shift variable, hence central—and on the elementary behaviour of the Smith normal form under direct sums. No unproven law and no external identification are used. The result is the rigorous half of a torsion detector: with no computation it says *where* a counterexample must live, for every cell and every prime dividing the degree. It does not say *what* rank a primitive block carries; that is a separate, validated-but-not-proven law and is deliberately excluded here. The torsion counterexample has not been found; every measured block is primitive. What this note contributes is the proof of confinement—the per-block elementary-divisor structure that Aljovin–Movasati–Villaflor explicitly left uncomputed.

How to read this. Section 1 fixes notation. Section 2 is the theorem and a self-contained proof; it may be read alone. Section 3 works one idempotent explicitly so the construction is concrete. Section 4 states precisely what the result does and does not give, and Section 5 records the machine verification of its hypotheses. The punchline, for a reader in a hurry: localization is proven; it tells you where a counterexample must be, not whether one exists.

1 Notation and setting

Fix a Fermat cell (n, m) with $n = 2d$ even, and a prime $p \mid m$. Over $\mathbb{F}_p$ the polynomial $t^m - 1$ (equivalently the relevant cyclotomic quotient φ) factors into coprime primary components. Grouping the $(t-1)$ -primary part as A and the remainder as B gives, by the Chinese Remainder Theorem, a splitting of the per-variable quotient ring

$$\mathbb{F}_p[t]/(\varphi) \cong A \times B, \quad \dim_{\mathbb{F}_p} A + \dim_{\mathbb{F}_p} B = m - 1.$$

Write $NV = n + 1$ for the number of variables. Tensoring the per-variable splitting over the NV variables decomposes the cohomology module M into *blocks*, indexed by which of the NV variables are placed in which primary component. For the two-component case $A \times B$ —the case in which every cell of the underlying campaign lives—there are 2^{NV} blocks, indexed by a binary mask of length NV . The general component count is not needed for the present result; the theorem below holds for any number of components.

Throughout, M denotes the integral cohomology module of the cell carrying the relevant $\mathbb{Z}$ -lattice structure, and "rank modulo p " refers to the $\mathbb{F}_p$ -rank of the reduced presentation. The CRT decomposition is taken over $\mathbb{F}_p$; the integral statement follows because the idempotents, being central, respect the lattice's block structure.

2 The theorem and its proof

Theorem 1 (Localization). *For every Fermat cell (n, m) and every prime $p \mid m$, the cohomology module M decomposes as a direct sum over the CRT blocks,*

$$M = \bigoplus_i e_i M,$$

where $\{e_i\}$ are the CRT idempotents. Consequently any torsion subgroup of M is confined to a single block—the unique block whose rank drops modulo p .

Proof. The CRT idempotents e_i effecting the splitting are polynomials in the shift variable t : they are the standard idempotents

$$e_i = \prod_{j \neq i} \frac{t - r_j}{r_i - r_j}$$

lifted from the factorization of the minimal polynomial of multiplication-by- t into coprime primary factors. (When a factor is not linear the same construction applies with the corresponding primary polynomial in place of $t - r_j$, obtained from a Bezout relation between the coprime factors; the resulting e_i is still a polynomial in t . See Section 3 for an explicit instance.)

A polynomial in t commutes with multiplication by t . Since the entire module structure of M is generated by the shift operator, each e_i commutes with the action and is therefore *central*. The idempotents are *orthogonal*, $e_i e_j = 0$ for $i \neq j$, and *complete*, $\sum_i e_i = 1$; these are the defining relations of the CRT decomposition.

A central, orthogonal, complete system of idempotents yields a direct-sum decomposition of the module,

$$M = \bigoplus_i e_i M,$$

into submodules, each invariant under the action. With respect to a basis adapted to this decomposition the presentation matrix of M is block-diagonal, one block per summand.

The Smith normal form of a block-diagonal matrix is the concatenation of the Smith normal forms of its diagonal blocks: the elementary divisors of the whole are exactly the union, with multiplicity, of the elementary divisors of the blocks. Since the torsion of M is precisely the part of the cokernel recorded by the non-unit elementary divisors, the torsion partitions across the blocks. Hence any torsion subgroup is localized to the block (or blocks) carrying a non-unit elementary divisor—equivalently, to the block whose $\mathbb{F}_p$ -rank is strictly less than its $\mathbb{Z}$ -rank, i.e. the block that drops rank modulo p . $\square$

Remark 1. The proof uses nothing beyond the idempotents being polynomials in the shift and the additivity of the Smith normal form under direct sums. In particular it requires no dictionary between these blocks and any external geometric construction, and no transfer of an external corollary. It holds verbatim for any number of coprime primary components, not only the two-component case.

3 A worked idempotent

To make the construction concrete, take the cell of degree $m = 6$ at the prime $p = 2$. Over $\mathbb{F}_2$,

$$\varphi(t) = \frac{t^6 - 1}{t - 1} = t^5 + t^4 + t^3 + t^2 + t + 1 = (t + 1)(t^2 + t + 1)^2,$$

so the $(t-1)$ -primary part is $A = (t+1)$ with $\dim_{\mathbb{F}_2} A = 1$ and the coprime cofactor is $B = (t^2+t+1)^2$ with $\dim_{\mathbb{F}_2} B = 4$ (note $1+4 = 5 = m-1$). Since $\gcd(A, B) = 1$, the extended Euclidean algorithm over $\mathbb{F}_2$ gives a Bezout relation

$$sA + uB = 1, \quad s = t^3 + t^2, \quad u = 1,$$

and the idempotents, reduced modulo φ , are

$$e_B = sA \bmod \varphi = t^4 + t^2, \quad e_A = uB \bmod \varphi = t^4 + t^2 + 1.$$

One checks directly, over $\mathbb{F}_2$ and modulo φ , that

$$e_A^2 = e_A, \quad e_B^2 = e_B, \quad e_A e_B = 0, \quad e_A + e_B = 1,$$

and each e_i , being a polynomial in t , commutes with multiplication by t —the centrality the proof uses. These are exactly the four conditions invoked in Theorem 1, here exhibited on an explicit pair. The same computation runs for any cell and prime; the verifier of Section 5 performs it across several.

4 Significance, and what the result does not give

What it gives. Theorem 1 is the rigorous half of a torsion detector. For every cell, with no computation and no unproven law, it pins the only place a counterexample to integral primitivity can live: the single block that drops rank modulo p . Every other block is torsion-free by the partition of elementary divisors. This converts an unbounded search over the whole cohomology lattice into the inspection of one identified block.

What it does not give. Localization says *where*, not *what*. It does not by itself supply the rank a primitive block should carry—that is a separate per-block rank law, which at present is strongly validated out of sample but not proven, and is therefore excluded from this note by design. Nor does the theorem constitute, on its own, a tested detector: a detector requires both the localization (proven here) and the value law (not proven). Conflating the two would dress this result above its evidence. The honest statement is: the location of any torsion is settled; the value that would betray it is not yet a theorem.

Scope. This concerns the integral Hodge conjecture for Fermat varieties—a specific, bounded problem. It is not the Clay Millennium (rational) Hodge conjecture and makes no claim on it. The torsion counterexample has not been found; every measured block in the campaign is primitive, consistent with the conjecture. What Theorem 1 adds to the literature is a proof of confinement: the per-block elementary-divisor structure that Aljovin–Movasati–Villaflor [2] tabulated the ambient lattice for but explicitly left uncomputed.

5 Reproducibility

The hypotheses of Theorem 1—that the constructed idempotents are central, orthogonal, idempotent, and complete—are verified byte-exact by a self-contained Python script that recomputes the idempotents from the factorization of $t^m - 1$ over $\mathbb{F}_p$ for several cells and checks each property directly:

`python3 verifiers/CENTRALITY_DETECTOR_VERIFIER.py`

The script constructs the idempotents for characteristics $p \in \{2, 3, 5\}$ and degrees $m \in \{6, 12, 15\}$, and confirms in each case that $e_i e_j = \delta_{ij} e_i$, that $\sum_i e_i = 1$, and that each e_i commutes with multiplication by t (centrality)—the worked case $m = 6, p = 2$ of Section 3 among them. These are exactly the conditions invoked in the proof; no value law and no external construction enter the verification.

References

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- [2] E. Aljovin, H. Movasati, R. Villafior, *Integral Hodge conjecture for Fermat varieties*. J. Symbolic Computation **95** (2019), 177–184. arXiv:1711.02628.